

have been ideas in science fiction for a century, but were first suggested by Johannes Kepler in a letter to Galileo in 1610 after he noticed that comets always had tails pointed away from the sun. This suggested that there was some sort of current that flowed out from the sun, and thus, could theoretically be harnessed with a large sail – much as humans have done using winds for centuries. His words, “Provide ships or sails adapted to the heavenly breezes, and there will be some who will brave even that void.”, came true almost exactly 400 years later. The Japanese launched IKAROS (Interplanetary Kite-craft Accelerated by Radiation Of the Sun) (also a play on Icarus, from Greek mythology) in May 2010. This was the first practical test of the solar sail idea, and it’s already achieved what it set out to do. Solar sails work, and IKAROS is now recognised as the smallest spacecraft flying between the planets of our precious solar system.

If research continues in this direction, it’s not hard to imagine huge sails pulling small spacecraft between the planets, leading the way in colonising the solar system. Thankfully, there’s plenty of space out there.

Better combustion

Almost a century later, and only incremental changes have been made to the ICE (internal combustion engine). Sure in a hundred years fuel efficiency has more than doubled, and horsepower is over 10 times greater, but would you be happy with so little improvement in any other area of life, given the time period? The idea is still pretty much the same – take a small amount of fuel, pressurize it, burn it (explode it?), use the energy to push a piston, which in turn pushes a crank shaft

and turns your wheels. What’s improved is that the amount of fuel burnt is lower, the power greater, the pressures higher and the wheels moving faster. However, auto mechanics or engineers from the very early days of the ICE are the only set of time travellers who will not look at the advancements made in their field as a magic trick, and that means we haven’t done enough.

The Scuderi group are working on improving the ICE to provide better efficiency. Currently, about two-thirds of fuel is wasted in ICEs and this is what Scuderi hopes to fix. You can read more about their efforts at <http://www.scuderigroup.com/technology/>, but the basic idea they propose is to separate the combustion and exhaust cylinders, and thus improve efficiency. While normal ICEs run in a four

bustion every cycle. Thus you have a four-cylinder engine that acts like a two stroke engine. The design also allows for some more creative thinking, in the sense that since the four cylinders are not identical in function, combustion and compression does not have to be a cyclical affair. Instead, say, when the car is cruising or coasting along, the compressed air can be routed to a storage tank, and used later to power the engine when needed, without sucking in more fuel. Theoretically, this makes it much more efficient, reduces emissions, and thus, is greener.

Invisible, invincible, inevitable...

That technology will integrate seamlessly into our lives and remain invisible, yet invincible is a given. The future has never been more exciting, and no, it’s not just smart phones that will improve your life, it’s services and hidden technologies that perhaps phones and apps will tap into that will really blow your future self’s mind. As technology gets smarter and better, the five areas we’ve mentioned in this article will come together seamlessly to improve efficiency. Future you (and your boss) will be so used to networks, that the idea of you being present only digitally will suffice. Improved identification will make our networks more secure, and reduce the instances of hacking and identity theft. Automation will enable you to offset the drudgery of the everyday and focus your brain on creativity and problem solving, and again networks can ensure that you can do this task from anywhere, and not necessarily in a stuffy office. Sensors will make the technology layer disappear all together, and become truly invisible, and transport will ensure that, although you can telecommute to work, when you go on holiday, you don’t ruin the environment, or better still, take a vacation on Mars, while your AI network does your work for you... one can dream!

Alas, space constraints always dictate that we prioritise, so if you think we missed something important, or would like to know more about certain aspects of the promising improvements mentioned here, don’t forget to drop us a line, and we promise to try and cover them in future issues or Fast Tracks. 

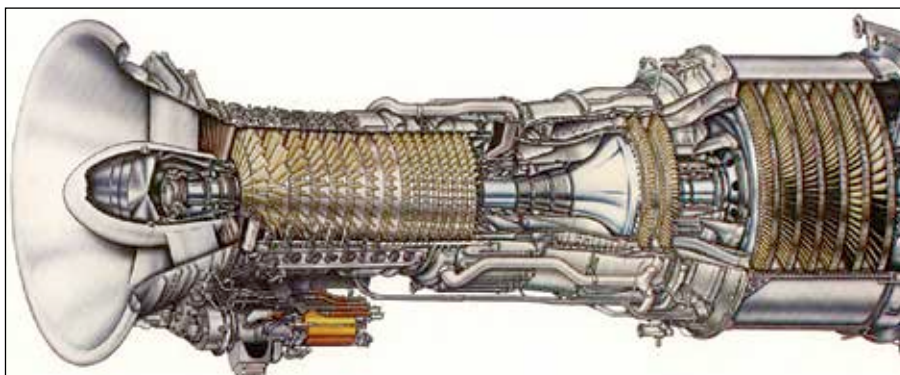
Power for the future



Gas-turbine engines have been here a long time and aircraft and ships have taken advantage of their efficiency and power

for a very long time now. Cars are now waking up to the potential of this technology as well. Head over to <http://digit.in/CSJuly5> to find out more about this. As a bonus, you get a sneak peek at the largest solar sail as well.

stage motion – intake of fuel, compression of fuel air mix, combustion (the power stroke) and finally exhaust – Scuderi aims to have two cylinders handling the compression and two others handling combustion. Instead of a power stroke (combustion) every two revolutions of the engine, the Scuderi engine will have com-



Gas turbines. The engines that will power the future?